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### **MAT FOR BASE PLATE WITH APERTURES**

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#### **Abstract**

An apparatus comprises a base plate including a plurality of apertures and a mat. At least a first set of apertures of the plurality of apertures are configured to receive a mounting foot from an accessory to be mounted on the base plate. The mat includes an attachment structure that is selectively securable to the base plate via one or more apertures of the plurality of apertures.

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#### **Background/Summary**

##### TECHNICAL FIELD

[0001] This disclosure relates generally to a mat that can be selectively secured to a base plate that is used to attach accessories within a cargo or work area of a vehicle.

## BACKGROUND

[0002] Vehicles transport various types of cargo. A user can, for example, rely on a vehicle to transport accessories, such as containers and tools, to and from a jobsite. The accessories that need to be transported can vary based on the tasks the user performs at the jobsite. Support surfaces in cargo or work areas may be configured in different ways to securely support the accessories.

## SUMMARY

[0003] An apparatus according to an exemplary aspect of the present disclosure includes, among other things: a base plate including a plurality of apertures, wherein at least a first set of apertures of the plurality of apertures are configured to receive a mounting foot from an accessory to be mounted on the base plate; and a mat including an attachment structure that is selectively securable to the base plate via one or more apertures of the plurality of apertures.

[0004] In a further non-limiting embodiment of the foregoing apparatus, the base plate is spaced from a bottom surface of a cargo area by a gap, and where the mat is selectively storable within the gap underneath the base plate.

[0005] In a further non-limiting embodiment of any of the foregoing apparatuses, the base plate comprises a first base plate positioned at a first vertical position relative to a bottom surface of a cargo area, and including a second base plate positioned at a second vertical position relative to the bottom surface of the cargo area that is above or below the first base plate, and wherein the mat is selectively securable to either the first base plate or the second base plate.

[0006] In a further non-limiting embodiment of any of the foregoing apparatuses, an upper surface of the mat has a first vertical height relative to a bottom surface of a cargo area at a first mat end and a second vertical height relative to the bottom surface of the cargo area at a second mat end that is greater than the first vertical height.

[0007] In a further non-limiting embodiment of any of the foregoing apparatuses, the cargo area extends from a passenger area to a rear of a vehicle, and wherein the first mat end is positioned at the rear of the vehicle.

[0008] In a further non-limiting embodiment of any of the foregoing apparatuses, the mat comprises a central area surrounded by an outer peripheral edge, and wherein the mat has a thickness in the central area that is greater than a thickness of the mat at the outer peripheral edge.

[0009] In a further non-limiting embodiment of any of the foregoing apparatuses, the mat includes one or more handles.

[0010] In a further non-limiting embodiment of any of the foregoing apparatuses, the mat is comprised of a high density polyethylene plastic.

[0011] In a further non-limiting embodiment of any of the foregoing apparatuses, the mat includes a persistent luminescent material.

[0012] In a further non-limiting embodiment of any of the foregoing apparatuses, the mat includes one or more mat openings that align with one or more apertures of the plurality of apertures in the base plate, and wherein the attachment structure comprises one or more clips associated with the one or more mat openings and that secure the mat to the base plate while leaving the one or more mat openings uncovered.

[0013] In a further non-limiting embodiment of any of the foregoing apparatuses, the attachment structure comprises one or more protrusions formed with the mat that extend into one or more apertures of the plurality of apertures in the base plate.

[0014] In a further non-limiting embodiment of any of the foregoing apparatuses, the mat includes one or more mat openings that align with one or more apertures of the plurality of apertures in the base plate, and wherein the attachment structure comprises one or more fasteners that are inserted through the one or more mat openings and into the one or more apertures to secure the mat to the base plate and cover the one or more mat openings.

[0015] In a further non-limiting embodiment of any of the foregoing apparatuses, the mat includes one or more mat openings that align with one or more apertures of the plurality of apertures in the

base plate, and wherein the attachment structure comprises: [0016] an insert is pressed into the one or more apertures, the insert including a center opening and compressible fingers that extend downwardly around the center opening; and [0017] a post with an enlarged head that is inserted into the center opening to press the compressible fingers outwardly to secure the mat to the base plate.

[0018] In a further non-limiting embodiment of any of the foregoing apparatuses, the first set of apertures comprise mounting foot apertures having a first size, and wherein the plurality of apertures include a second set of apertures that comprise locking apertures through which the mat can be fixed to the base plate, wherein the locking apertures have a second size that is different than the first size.

[0019] A method according to an exemplary aspect of the present disclosure includes, among other things: providing a plurality apertures in a base plate, wherein at least a first set of apertures of the plurality of apertures are configured to receive a mounting foot from an accessory to be mounted on the base plate; and selectively attaching an attachment structure of a mat to the base plate via one or more apertures of the plurality of apertures.

[0020] In a further non-limiting embodiment of the foregoing method, the method includes: storing the mat in an open area underneath the base plate; pulling the mat out of the open area; and securing the mat to an upper surface of the base plate.

[0021] In a further non-limiting embodiment of any of the foregoing methods, the method includes: vacuum forming the mat from a high density polyethylene plastic material.

[0022] In a further non-limiting embodiment of any of the foregoing methods, the method includes: installing the mat within a cargo area of a vehicle; and positioning a portion of the mat near a passenger area at one end of the cargo area to be at a greater height relative to a portion of the mat in the cargo area near a rear of a vehicle.

[0023] In a further non-limiting embodiment of any of the foregoing methods, the method includes: forming a central area of the mat to have a greater thickness than a peripheral edge portion of the mat.

[0024] In a further non-limiting embodiment of any of the foregoing methods, the method includes: forming the attachment structure as one or more protrusions in the mat, and inserting the one or more protrusions into one or more apertures of the plurality of apertures in the base plate; or forming one or more mat openings in the mat that align with one or more apertures of the plurality of apertures in the base plate, and wherein the attachment structure comprises a plurality of resilient members that are initially compressed during insertion through the one or more apertures and subsequently expand after insertion to secure the mat to the base plate.

[0025] The embodiments, examples and alternatives of the preceding paragraphs, the claims, or the following description and drawings, including any of their various aspects or respective individual features, may be taken independently or in any combination. Features described in connection with one embodiment are applicable to all embodiments, unless such features are incompatible.

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## Description

### BRIEF DESCRIPTION OF THE FIGURES

[0026] The various features and advantages of the disclosed examples will become apparent to those skilled in the art from the detailed description. The figures that accompany the detailed description can be briefly described as follows:

[0027] FIG. 1 illustrates a perspective view of a vehicle cargo bed equipped with a base plate that can be used to secure an accessory.

[0028] FIG. 2 illustrates a top view of a mat that can be supported on the base plate according to an exemplary aspect of the present disclosure.

[0029] FIG. 3 illustrates a top view of the base plate supporting the mat.

[0030] FIG. 4A illustrates a perspective view of a mat supported with a cargo bed similar to that of FIG. 3.

[0031] FIG. 4B is a perspective view of the mat supported on an upper base plate.

[0032] FIG. 4C is a perspective view of the mat supported on a lower base plate.

[0033] FIG. 5 is a section view of one example attachment structure for attaching the mat to the base plate.

[0034] FIG. 6 is a perspective view of another example attachment structure for attaching the mat to the base plate.

[0035] FIG. 7 is a perspective view of another example attachment structure for attaching the mat to the base plate.

[0036] FIG. 8A is an exploded view of another example attachment structure for attaching the mat to the base plate.

[0037] FIG. 8B is an initial installation position of the attachment structure of FIG. 8A.

[0038] FIG. 8C is a final installation position of the attachment structure of FIG. 8A.

[0039] FIG. 8D is a top view of FIG. 8C.

[0040] FIG. 9 is a schematic illustration of a mat being position higher near a front of a cargo area than at a rear of the cargo area.

[0041] FIG. 10 is a schematic illustration of a mat have a smaller thickness at a peripheral edge and a greater thickness at a center portion of the mat.

#### DETAILED DESCRIPTION

[0042] This disclosure details a mat that is selectively securable to a base plate that is used with an accessory attachment system for a vehicle. The base plate provides an attachment interface comprising a plurality of apertures for securing the accessories in place within a cargo or work area. There are often situations where it would be useful for the vehicle owner to be able to cover the base plates/apertures with a protective structure such as a mat, for example. The subject disclosure provides mat that can be quickly and efficiently attached to a base plate to provide a continuous and substantially uninterrupted support surface.

[0043] With reference to FIGS. 1-3, a vehicle cargo area 10 has a first end adjacent a passenger compartment 12 and a second end that is at a rear portion 16, e.g., a tailgate, closure panel, etc., of a vehicle. The cargo area 10 has a floor 18. In this example, the vehicle is a pickup truck. However, the vehicle could be another type of vehicle in another example, such as a car, van, sport utility vehicle, etc. The example vehicle could also be a traditional internal combustion engine powered vehicle or an electrified vehicle, such as a battery electric vehicle (BEV) or plug-in hybrid electric vehicle (PHEV) for example.

[0044] In the exemplary embodiment, one or more base plates 20 are used to support one or more accessories and/or modules 22. In one example, the base plates 20 are secured directly to the floor 18 of the cargo area 10 by fastening, welding, etc., at a first connection interface. In another example shown in FIG. 4B, the base plates 20 can be supported for sliding movement within the cargo area 10 on tracks, rails 24, or other types of sliding structures. An example of such a system with sliding movement is found in application Ser. No. 17/886,663, which was filed on Aug. 12, 2022, which is assigned to the assignee of the subject application, and which is herein incorporated by reference. The accessories and/or modules 22 can be secured to the vehicle by engaging one or more of the base plates 20 via a second connection interface between the accessory 22 and the base plate 20. The accessories/modules 22 can comprise a lockable storage box that holds tools, a container, a refrigerator, etc.

[0045] As shown in FIG. 1, the base plate 20 includes a plurality of openings or apertures 26. An example accessory 22 comprises a lockable container or box 22 that includes one or more mounting feet 28. The box 22 can engage the base plate 20 using an attachment system comprising the plurality of apertures 26 and the mounting feet 28 that cooperate to provide a mechanical

connection interface. The base plate **20** and the apertures **26** are spaced upwardly from the floor **18** by an open gap such that the feet **28** from the accessory **22** can be easily inserted into the apertures **26** in the base plate **20** to attach the accessory **22** to the base plate **20**.

[0046] In one example, a mat **30** (FIG. 2) is selectively securable to the base plate **20** to cover the base plate **20** and apertures **26**. In one example, the mat **30** comprises a generally rectangular shaped body having a central area **32** surrounded by an outer peripheral edge **34**. The mat **30** includes an attachment interface **36** that is used to selectively secure the mat **30** to the base plate **20** via one or more apertures of the plurality of apertures in the base plate **20**.

[0047] In one example, the apertures **26** that are formed within the base plate **20** comprise a plate attachment interface for accessories **22**. However, the base plate **20** can also include additional apertures to provide other attachment features or power connection features. In one example, the base plate **20** includes a plurality of connection and/or lock apertures **38** that interact with the attachment interface **36** of the mat **30**. These lock apertures **38** are provided at multiple locations on the base plate **20**, and are formed in a desired pattern relative to the apertures **26** for the mounting feet **28** of the accessories **22**. In one example, the feet apertures **26** and lock apertures **38** have different sizes and/or shapes. In one example, the lock apertures **38** are smaller in size than the feet apertures **26**.

[0048] As discussed above, there is an open area **40** underneath the base plate **20**. In one example, the mat **30** is selectively storable within this open area **40** when not needed. When the mat **30** is needed, the mat **30** can be easily pulled out of the open area **40** via one or more handles **42**, for example, and placed on an upper surface of the base plate **20**. The mat **30** can then be secured to the base plate **20** via the attachment interface **36**.

[0049] In one example, the handles **42** comprise openings **44** integrally formed in the mat **30** near a front edge **46** of the mat **30**. The openings **44** provide for a grip portion **48** that can be grasped by a user to pull the mat **30** out of the open area **40** and/or to pull the mat **30** upwardly away from the base plate **20** for detachment purposes.

[0050] When in the stowed/storage position, the mat **30** can rest on the floor **18** of the cargo area **10** as shown in FIG. 4A. Optionally, a plurality of base plates **20** can be mounted within with the tracks/rails **24** to provide different support surface heights. FIG. 4B shows an example where the mat **30** is supported on a base plate **20** that has been slid out from a lower position **50** and slid into a higher position **52** (see arrow S).

[0051] FIG. 4C shows an example where the base plate comprises a first base plate **20a** positioned at the lower position **50**, i.e., a first vertical position relative to a bottom surface of the cargo area, and a second base plate **20b** is provided that is positioned at the higher position **52**, i.e., a second vertical position relative to the bottom surface of the cargo area that is above the first base plate **20a**. The mat **30** is selectively securable to either the first base plate **20a** or the second base plate **20b**. Optionally, each base plate **20** could be covered by a mat **30**. In the example shown in FIG. 4C, the mat **30** is attached to the lower base plate **20a**.

[0052] FIG. 5 shows one example of the attachment structure **36**. In this example, the attachment structure **36** comprises one or more protrusions **62** formed with the mat **30**. In one example, there are at least four protrusion **62**, e.g., one spaced inward from each corner of the mat **30**, that extend into corresponding lock apertures **38** in the base plate **20**. The protrusions **62** can be integrally formed with the mat **30** or separately attached thereto. Further, fewer or additional protrusions can be used as needed.

[0053] FIG. 6 shows another example of the attachment structure **36**. In this example, the attachment structure **36** comprises one or more clips **64**. In one example, there are at least four clips **64**, e.g., one spaced inward from each corner of the mat **30**, that extend into corresponding lock apertures **38** in the base plate **20**. Fewer or additional clips **64** can be used as needed. In this example, the attachment structure **36** further comprises mat openings **66** that align with the lock apertures **38** of the base plate **20**. In this example configuration, the clips **64** are associated with the

mat openings **66** and secure the mat **30** to the base plate **20** while leaving the mat openings **66** uncovered. The base plate lock openings **38** may also be accessible in this configuration.

[0054] In one example, the clips **64** comprise at least two compressible tabs or fingers **68** that are connected to each other via a U-shaped body portion **70**, and a second pair of compressible tabs or fingers **72** are outward of the first pair of compressible tabs or fingers **68**. The second pair of compressible tabs or fingers **72** are underneath the mat **30** and are associated with the base plate **20**, and the first pair of compressible tabs or fingers **68** extend through the mat openings **66**. The first pair of compressible tabs or fingers **68** are compressed inwardly toward each other to fit through the openings **66** and then expand outwardly to secure the mat **30** to the base plate **20**.

[0055] FIG. 7 shows another example of the attachment structure **36**. In this example, the attachment structure **36** comprises one or more fasteners **74**. In one example, there are at least four fasteners **74**, e.g., one spaced inward from each corner of the mat **30**, that extend into corresponding lock apertures **38** in the base plate **20**. Fewer or additional fasteners **74** can be used as needed. In this example, the attachment structure **36** further comprises the mat openings **66** that align with the lock apertures **38** of the base plate **20**. In this example configuration, the fasteners **74** are associated with the mat openings **66** and secure the mat **30** to the base plate **20** while leaving the mat openings **66** and lock openings **38** covered. In one example, the fasteners **74** comprise a flat base **76** with a center extension comprising a plurality of compressible tabs or fingers **78** that are inserted through the mat openings **66** and into the lock apertures **38** to secure the mat **30** to the base plate **20** such that the base **76** covers the mat openings **66**. The compressible tabs or fingers **78** compress inwardly during insertion and expand outwardly once inserted. A user can grip the base **76** to pull the fastener **74** away from the mat, or the handles **42** can be used to lift the fasteners **74** out.

[0056] FIGS. 8A-8D show another example of the attachment structure **36**. In this example, the attachment structure **36** comprises one or more inserts **80** and posts **82**. In one example, there are at least four sets of inserts and posts, e.g., one set spaced inward from each corner of the mat **30**, that extend into corresponding lock apertures **38** in the base plate **20**. Fewer or additional sets of inserts and posts can be used as needed. In this example, the attachment structure **36** further comprises the mat openings **66** that align with the lock apertures **38** of the base plate **20**. In this example configuration, the inserts **80** and posts **82** are associated with the mat openings **66** and secure the mat **30** to the base plate **20** while leaving the mat openings **66** and lock openings **38** covered.

[0057] In one example, the insert **80** is first pressed into the corresponding lock apertures **38**. The insert **80** includes a center opening **84** and compressible fingers **86** that extend downwardly around the center opening **84**. In one example, the post **82** comprises an enlarged head **88** with a reduced linear portion **90** extending away from the head **88**. The linear portion **90** is inserted into the center opening **84** to press the compressible fingers **86** outwardly to secure the mat **30** to the base plate **20**. In this example, a tool (not shown) may be required to pull out the post **82**.

[0058] As shown schematically in FIG. 9, an upper surface **54** of the mat **30** has a first vertical **V1** height relative to a bottom surface **56** of the cargo area **10** at a first mat end **58** and a second vertical height **V2** relative to the bottom surface **56** of the cargo area **10** at a second mat end **60** that is greater than the first vertical height **V1**. This allows for drainage to be managed in a desired direction. In one example, the cargo area **10** extends from the passenger compartment **12** to the rear portion **16**, e.g., a tailgate, closure panel, etc., of the vehicle, and the first mat end **58** is positioned at the rear portion **16** of the vehicle such that fluids can drain away from the passenger compartment **12**.

[0059] As discussed above, the mat **30** comprises a central area **32** that is surrounded by an outer peripheral edge **34**. In one example, the mat **30** has a thickness **T1** in the central area **32** that is greater than a thickness **T2** of the mat **30** at the outer peripheral edge **34** as shown in FIG. 10. This configuration can facilitate sealing. For example, the mat **30** seals at the edges **34** by being stretched more at the periphery to make these edges **34** more flexible by having a thinner wall

thickness. In one example, the cross-section of the mat **30** comprises a wave pattern to give a thinner wall thickness as shown in FIG. **10**.

[0060] In one example, the mat **30** is comprised of low friction material, e.g., a high density polyethylene (HDPE) plastic or other similar materials. Further, the mat **30** have varying degrees of softness/hardness. In one example, the mat **30** is vacuum formed; however, other forming processes could also be used. The use of a low friction plastic material allows the mat **30** to slide out when pulled by the handles **42** at the front edge **46** of the mat **30**.

[0061] In one example, the mat includes a persistent luminescent material, e.g., glow-in-the-dark phosphors. In one example, this material is molded in combination with the plastic material. This can provide lighting underneath the base plate **20** when the mat **30** is in a stowed position. The phosphor material is charged during the day from light coming through openings in the base plate and then can glow at night to provide a supplemental light source.

[0062] The subject disclosure provides for a mat that goes into a cargo bed underneath the base plate. The mat protects the bed from water and debris, is easy to be pulled out to be cleaned, and can be selectively locked in place on the base plate. Handles are provided for pulling out the mat which can be comprised of a low friction plastic to further facilitate sliding movement. Water management, e.g., drainage control, is also provided as the mat is higher in near the cabin than at the rear of the vehicle so water can run out. Further, the mat is thinned out on the edges so a friction fit sealing against the vehicle one bed can be created.

[0063] The preceding description is exemplary rather than limiting in nature. Variations and modifications to the disclosed examples may become apparent to those skilled in the art that do not necessarily depart from the essence of this disclosure. Thus, the scope of protection given to this disclosure can only be determined by studying the following claims.

## Claims

1. An apparatus, comprising: a base plate including a plurality of apertures, wherein at least a first set of apertures of the plurality of apertures are configured to receive a mounting foot from an accessory to be mounted on the base plate; and a mat including an attachment structure that is selectively securable to the base plate via one or more apertures of the plurality of apertures.
2. The apparatus of claim 1, wherein the base plate is spaced from a bottom surface of a cargo area by a gap, and wherein the mat is selectively storable within the gap underneath the base plate.
3. The apparatus of claim 1, wherein the base plate comprises a first base plate positioned at a first vertical position relative to a bottom surface of a cargo area, and including a second base plate positioned at a second vertical position relative to the bottom surface of the cargo area that is above or below the first base plate, and wherein the mat is selectively securable to either the first base plate or the second base plate.
4. The apparatus of claim 1, wherein an upper surface of the mat has a first vertical height relative to a bottom surface of a cargo area at a first mat end and a second vertical height relative to the bottom surface of the cargo area at a second mat end that is greater than the first vertical height.
5. The apparatus of claim 4, wherein the cargo area extends from a passenger area to a rear of a vehicle, and wherein the first mat end is positioned at the rear of the vehicle.
6. The apparatus of claim 1, wherein the mat comprises a central area surrounded by an outer peripheral edge, and wherein the mat has a thickness in the central area that is greater than a thickness of the mat at the outer peripheral edge.
7. The apparatus of claim 1, wherein the mat includes one or more handles.
8. The apparatus of claim 1, wherein the mat is comprised of a high density polyethylene plastic.
9. The apparatus of claim 1, wherein the mat includes a persistent luminescent material.
10. The apparatus of claim 1, wherein the mat includes one or more mat openings that align with one or more apertures of the plurality of apertures in the base plate, and wherein the attachment

structure comprises one or more clips associated with the one or more mat openings and that secure the mat to the base plate while leaving the one or more mat openings uncovered.

**11.** The apparatus of claim 1, wherein the attachment structure comprises one or more protrusions formed with the mat that extend into one or more apertures of the plurality of apertures in the base plate.

**12.** The apparatus of claim 1, wherein the mat includes one or more mat openings that align with one or more apertures of the plurality of apertures in the base plate, and wherein the attachment structure comprises one or more fasteners that are inserted through the one or more mat openings and into the one or more apertures to secure the mat to the base plate and cover the one or more mat openings.

**13.** The apparatus of claim 1, wherein the mat includes one or more mat openings that align with one or more apertures of the plurality of apertures in the base plate, and wherein the attachment structure comprises: an insert is pressed into the one or more apertures, the insert including a center opening and compressible fingers that extend downwardly around the center opening; and a post with an enlarged head that is inserted into the center opening to press the compressible fingers outwardly to secure the mat to the base plate.

**14.** The apparatus of claim 1, wherein the first set of apertures comprise mounting foot apertures having a first size, and wherein the plurality of apertures include a second set of apertures that comprise locking apertures through which the mat can be fixed to the base plate, wherein the locking apertures have a second size that is different than the first size.

**15.** A method comprising: providing a plurality of apertures in a base plate, wherein at least a first set of apertures of the plurality of apertures are configured to receive a mounting foot from an accessory to be mounted on the base plate; and selectively attaching an attachment structure of a mat to the base plate via one or more apertures of the plurality of apertures.

**16.** The method of claim 15, including: storing the mat in an open area underneath the base plate; pulling the mat out of the open area; and securing the mat to an upper surface of the base plate.

**17.** The method of claim 15, including vacuum forming the mat from a high density polyethylene plastic material.

**18.** The method of claim 15, including installing the mat within a cargo area of a vehicle, and positioning a portion of the mat near a passenger area at one end of the cargo area to be at a greater height relative to a portion of the mat in the cargo area near a rear of a vehicle.

**19.** The method of claim 15, including forming a central area of the mat to have a greater thickness than a peripheral edge portion of the mat.

**20.** The method of claim 15, including: forming the attachment structure as one or more protrusions in the mat, and inserting the one or more protrusions into one or more apertures of the plurality of apertures in the base plate; or forming one or more mat openings in the mat that align with one or more apertures of the plurality of apertures in the base plate, and wherein the attachment structure comprises a plurality of resilient members that are initially compressed during insertion through the one or more apertures and subsequently expand after insertion to secure the mat to the base plate.

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